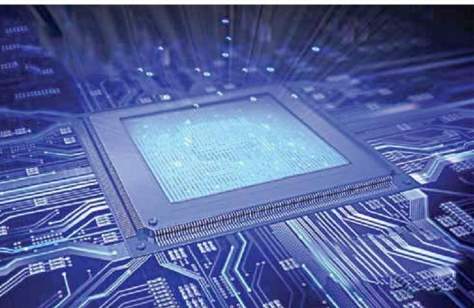


FPGA Vs ASIC for High Volume

The choice of ASIC vs FPGA for volume production is on case-to-case basis. However, what if there were a way to get the best of both the worlds?



DILIN ANAND

A field-programmable gate array (FPGA) can be purchased off-the-shelf and programmed by the user, whereas an application-specific integrated circuit (ASIC) is manufactured to a customer's specification. This distinction has not changed since the dawn of both technologies. However, the performance offered by these two technologies, and the economics that go behind them, have changed considerably over the years. The time is ripe to discuss again and see what can be done about this conundrum.

Time-to-market and configurability

The configurability of an FPGA is its most prized asset. Its ability to rapidly implement or reprogram the logic for a specific feature, or to modify the func-

tionality that was previously instilled in it, is why designers run after it.

"Even if a vendor has new features to add at a later stage in the released product, he still has the freedom to decide whether to implement that feature in software or hardware based on applicability. Time-to-market for handling change-requests in FPGA is much less than in ASICs," explains Manisha Mankar, architect—digital design, Robert Bosch Engineering and Business Solutions.

Indeed, this is one of the primary reasons why designers are opting for FPGAs.

"There are two key factors driving the demand for FPGAs today: The programmable imperative, i.e., the inherent capability of an FPGA that allows it to be tailored to the needs of the customer and the insatiable bandwidth requirements of the mobile generation today. Given this, FPGAs are increasingly replac-

ing ASICs and ASSPs for more and more applications across different verticals such as telecommunications, aerospace, medical, automotive and industrial to name a few," adds Neeraj Varma, director-sales, Xilinx India.

Of course, an FPGA would not be able to achieve the smaller form factor, lower power consumption or lower unit cost of a similarly functioning ASIC.

Performance and power efficiency

While FPGAs used to be selected for lower-speed, less complex or volume designs in the past, today's FPGAs easily push the performance barrier. With increase in logic density and other features such as embedded processors, DSP blocks and high-speed serial at low price points, FPGAs are an interesting proposition. The industry has shown that high-end FPGAs are growing in volume, handling high-speed applications and complex designs.

Earlier, FPGAs were viable only for prototyping or low-density applications. Now they meet the needs of very high-volume applications such as consumer products and other moderate-volume high-density applications as well.

Vijayakumar Regupathi, architect-digital design, Robert Bosch Engineering and Business Solutions, adds, "FPGA devices now look like system-on-chips (SoC) with embedded processors, signal processing block, embedded memory interface controller, multi-gigabit transceivers, improved performance and a broad choice of IPs available from FPGA vendors and/or third parties.